Ricci-energy selection, scalar coherence, and Einstein defect

1. The question this note resolves

The preceding Kähler calculation identified the limiting relative Einstein defect of a constant-scalar-curvature blow-up sequence. Here we remove the Kähler assumption and describe **every** sequence approaching the sharp Ricci-energy bound. The remaining term is the scalar variance. An actual conformal construction shows that this term cannot be dropped, even on a fixed manifold admitting Kähler–Einstein metrics and with zero primitive period cost. Conversely, choosing the negative Yamabe representative removes this term without a spectral-gap assumption.

The finite estimates below follow from classical Seiberg–Witten and Chern–Weil inequalities. They do not require the project's integral calibration lemma, its second-order estimates, or the Round21 spectral criterion. This is a new deduction recorded in this research package, not a claim of priority over the literature.

All manifolds are closed, connected, oriented, and four-dimensional. Ricci norms are ordinary tensor norms; Weyl norms are endomorphism norms on two-forms. Write

$$T = 2\chi + 3\tau, \quad V = \mathrm{Vol}(g), \quad S_2 = \int s^2, \quad R = \int |\mathrm{Ric}|^2, \quad P = \int |\mathrm{Ric}^\circ|^2, \quad L = \int (s - \bar{s})^2. \tag{1}$$

Integrals use the current metric.

2. Precise topological input and the finite theorem

Fix numbers $A > 0$, $d \geqslant 0$, with $T = A - d$. For the metric in question, assume an actual polarized monopole Spin^c structure whose class a satisfies $p = (a_g^+)^2 \geqslant A$. Its equations must be solvable throughout the conformal class, so that the classical scalar and scalar–Weyl inequalities apply. A formal intersection vector is not enough.

The class may depend on g. No class-square upper bound, scalar band, Sobolev constant, injectivity-radius bound, or diameter bound is assumed.

In the standard application,

$$M = X \# k\overline{\mathbb{CP}}^2 \# m(S^1 \times S^3), \qquad A = c_1^2(X) > 0, \quad d = k + 4m, \tag{2}$$

where X is a minimal complex surface of general type. The polarized class selection and its $b_+ = 1$ interpretation are as in LeBrun's Proposition 3.2 and the application closure. LeBrun's Theorem 4.3 identifies the lower bound below as the actual all-metrics infimum in this application.

Put

$$C = 8\pi^2 A, \quad H = 8\pi^2 d, \quad J = R - (C + H). \tag{3}$$

Theorem 1 (finite energy-selection bound). Under the stated input, $J \geqslant 0$, and

$$\boxed{H - \sqrt{J(C + J)} \leqslant P \leqslant H + J.} \tag{4}$$

The lower bound may of course be replaced by its positive part. With

$$\mathscr{E}(g) = \inf_{\lambda \neq 0} \frac{\int |\mathrm{Ric} - \lambda g|^2}{\lambda^2 V}, \tag{5}$$

one has the exact formula

$$\mathscr{E}(g) = \frac{4P + L}{R} \tag{6}$$

and consequently

$$\boxed{-\frac{4\sqrt{J(C+J)}}{C+H+J} \leqslant \mathscr{E}(g) - \frac{4H+L}{C+H+J} \leqslant \frac{4J}{C+H+J}.} \tag{7}$$

Thus the Einstein defect is not determined by absolute Ricci energy alone. Near its minimum, its two contributions are the topological quantity $4H$ and the scalar variance L.

Proof

Define the classical LeBrun remainder

$$D = \frac{1}{4\pi^2}\int\left(\frac{s^2}{24} + \frac{1}{2}|W^+|^2\right) - \frac{p}{2}. \tag{8}$$

The required classical inputs are

$$S_2 \geqslant 32\pi^2 p, \qquad \int(-s + \sqrt{6}|W^+|)^2 \geqslant 72\pi^2 p, \qquad D \geqslant 0. \tag{9}$$

These are the scalar estimate, Theorem 2.3, and Proposition 4.1 of LeBrun, arXiv:math/0003068v1. No assertion about the equality set of a different curvature functional is used.

Chern–Weil gives the exact identity

$$R = 8\pi^2(2p - T) + 32\pi^2 D, \qquad J = 16\pi^2(p - A) + 32\pi^2 D. \tag{10}$$

In particular

$$0 \leqslant p - A \leqslant J/(16\pi^2), \qquad 0 \leqslant D \leqslant J/(32\pi^2). \tag{11}$$

Let

$$q = -s + \sqrt{6}|W^+|, \quad t = s + 2\sqrt{6}|W^+|, \quad \delta = 144\pi^2 D, \quad \Delta = \int q^2 - 72\pi^2 p, \quad U = \int t^2. \tag{12}$$

Here q need not be positive. The unsigned inequality in (9) suffices. Pointwise expansion and integration yield

$$\delta = \Delta + \tfrac{1}{2}U, \qquad \Delta \geqslant 0, \qquad s = (t - 2q)/3. \tag{13}$$

For the scalar surplus $B_p = S_2 - 32\pi^2 p \geqslant 0$, this implies

$$9B_p = 4\delta - U - 4\int qt. \tag{14}$$

Cauchy–Schwarz and $U \leqslant 2\delta$ give

$$B_p \leqslant \frac{4}{9}\left[\delta + \sqrt{2\delta(72\pi^2 p + \delta)}\right]. \tag{15}$$

Using (11), $\delta \leqslant 9J/2$, and the convenient bound $72\pi^2 p + \delta \leqslant 72\pi^2 A + 9J$, we obtain

$$\boxed{0 \leqslant B_p \leqslant 2J + 4\sqrt{J(C+J)}.} \tag{16}$$

There is room to sharpen these numerical constants; their optimality is not asserted.

Since $P = R - S_2/4$, (10) also gives

$$P = H + J - 8\pi^2(p - A) - B_p/4. \tag{17}$$

The upper bound in (4) follows immediately. For the lower bound, (11) and (16) give

$$P - H \geqslant J - J/2 - \left(J/2 + \sqrt{J(C+J)}\right) = -\sqrt{J(C+J)}. \tag{18}$$

Finally, expansion in $1/\lambda$ yields

$$\mathscr{E} = 4 - \frac{(\int s)^2}{VR} = \frac{4P + L}{R}. \tag{19}$$

Here $R \geqslant C > 0$. If $\int s = 0$, the infimum is 4, approached as $|\lambda| \to \infty$; otherwise the minimizer is $\lambda = R/\int s$. This covers both cases without falsely treating a Ricci-flat metric as zero relative defect. Substitution proves (7).

3. All minimizing sequences, not only Kähler sequences

Corollary 2. In the setting of Theorem 1, if $J(g_j) \to 0$, then

$$p_j \to A, \quad D_j \to 0, \quad \int s_j^2 \to 32\pi^2 A, \quad P_j \to 8\pi^2 d. \tag{20}$$

Moreover

$$\boxed{\mathscr{E}(g_j) = \frac{4d}{A+d} + \frac{L_j}{8\pi^2(A+d)} + o(1).} \tag{21}$$

In particular

$$\liminf \mathscr{E}(g_j) \geqslant \frac{4d}{A+d}, \tag{22}$$

and $\mathscr{E}(g_j) \to 4d/(A+d)$ is equivalent to $L_j \to 0$. For $d = 0$, vanishing relative Einstein defect is equivalent to vanishing scalar variance along such a sequence.

Indeed (11) and (16) give (20). Since $0 \leqslant L_j \leqslant \int s_j^2$, the variance is bounded; (7) therefore gives (21). This assertion imposes no Kähler condition, no constant-scalar-curvature condition, and no geometric compactness hypothesis.

For varying topologies the uniform formulation is (7). For example, write $r = d/A$, $j = J/(8\pi^2 A)$, and $\ell = L/(8\pi^2 A)$. Then the difference between $\mathscr{E}$ and $(4r+\ell)/(1+r+j)$ lies between $-4\sqrt{j(1+j)}/(1+r+j)$ and $4j/(1+r+j)$. Small *relative* energy excess, rather than small unnormalized excess, is the appropriate hypothesis when A varies.

A direct scalar-coherence criterion on the original metric

For an ordinary monopole solution, put $\rho = |\Phi|^2$ and let $\mathscr{F}$ be its real determinant curvature. The integrated spinor identity and Hodge projection give

$$B_p = \|s + \rho\|_2^2 + 8\|\nabla\Phi\|_2^2 + 8\|\mathscr{F}^+ - (\mathscr{F}^+)_H\|_2^2. \tag{22a}$$

This retains the harmonic-projection loss in the classical scalar inequality. It also follows from the Round17 identity by subtracting the harmonic anti-self-dual contribution.

If $s \geqslant -S$, the maximum principle yields $0 \leqslant \rho \leqslant S$. Since $|d\rho|^2 \leqslant 4S|\nabla\Phi|^2$, Poincaré gives

$$\sqrt{L} \leqslant \|s + \rho\|_2 + \sqrt{4S/\lambda_1}\,\|\nabla\Phi\|_2.$$

Joint Cauchy–Schwarz in the two displayed losses of (22a) proves

$$\boxed{L \leqslant \left(1 + \frac{S}{2\lambda_1}\right) B_p.} \tag{22b}$$

Thus bounded S/λ_1 is one sufficient condition for scalar coherence along a minimizing sequence. It requires no weighted equations, strict negative scalar band, or prior bound on Ricci error. The stronger weighted variance estimate in the companion converse theorem instead controls L linearly by D, under its stated negative-band assumptions. Neither argument supplies a spectral gap for an arbitrary degenerating family.

4. Removing scalar variance by conformal normalization

The variance in (21) is essential on the original metric. It can nevertheless be removed by a mathematically legitimate change of representative.

Corollary 3 (Yamabe representatives of all minimizing sequences). In the standard application (2), or whenever Theorem 1's input holds in the conformal class, let $\widehat{g}$ be its negative constant-scalar- curvature Yamabe representative. Then

$$\hat{J} \leqslant J, \qquad R(g) - R(\widehat{g}) = \tfrac{1}{3}\big(S_2(g) - S_2(\widehat{g})\big), \qquad P(g) - P(\widehat{g}) = \tfrac{1}{12}\big(S_2(g) - S_2(\widehat{g})\big). \tag{23}$$

Consequently every Ricci-energy-minimizing sequence can be replaced, within its own conformal classes, by one satisfying

$$\boxed{\mathscr{E}(\widehat{g}_j) \longrightarrow \frac{4d}{A + d}.} \tag{24}$$

For $d = 0$ there is also the finite bound

$$\boxed{\mathscr{E}(\widehat{g}) \leqslant \frac{4J}{8\pi^2 A + J}.} \tag{25}$$

Proof of the conformal step

A nonzero actual monopole projection rules out nonnegative scalar curvature in this conformal class by the spinor Weitzenböck formula. The Yamabe theorem therefore provides a representative with negative constant scalar curvature. Four-dimensional Hodge star on two-forms, p, and $\int |W^+|^2$ are conformally invariant.

For completeness, the scalar-energy minimization in a negative conformal class has a direct verification. Rescale $\widehat{g}$ to $s_{\widehat{g}} = -b < 0$, and write $g = u^2\widehat{g}$. With $\Delta = \operatorname{div}\nabla$, conformal covariance gives

$$\int s_g^2 \, d\mu_g = \int \left(-b - 6u^{-1}\Delta_{\widehat{g}} u\right)^2 d\mu_{\widehat{g}}. \tag{26}$$

Integration by parts yields

$$\int u^{-1}\Delta u = \int u^{-2}|du|^2 \geqslant 0. \tag{27}$$

Expanding (26) shows $S_2(g) \geqslant b^2 V_{\widehat{g}} = S_2(\widehat{g})$. The Chern–Weil identity

$$R = 4\int |W^+|^2 + \tfrac{1}{3}S_2 - 8\pi^2 T \tag{28}$$

and $P = R - S_2/4$ prove (23). The lower bound $R(\widehat{g}) \geqslant C + H$ remains valid for the same polarization and actual class, so $0 \leqslant \hat{J} \leqslant J$. Now (21) with $\widehat{L} = 0$ proves (24). When $d = 0$, (4) gives $\widehat{P} \leqslant \hat{J}$, whence

$$\mathscr{E}(\widehat{g}) = \frac{4\widehat{P}}{C + \hat{J}} \leqslant \frac{4\hat{J}}{C + \hat{J}} \leqslant \frac{4J}{C + J}. \tag{29}$$

This does not estimate the distance between g and $\widehat{g}$. The conformal scale-splitting example shows that such an estimate cannot be inferred from $J \to 0$. Nor does an integral estimate for $\widehat{g}$ give a pointwise Ricci estimate.

Round14 already carried out Yamabe normalization for a particular shear construction. The assertion here concerns arbitrary input conformal classes and all Ricci-energy-minimizing sequences; it does not count the act of Yamabe normalization as a new method.

5. An explicit energy exclusion for all Einstein candidates

If a metric in this topological range is Einstein, then $P = 0$. Theorem 1 forces

$$J(C + J) \geqslant H^2, \qquad \boxed{J \geqslant \frac{\sqrt{C^2 + 4H^2} - C}{2} = 4\pi^2\Big(\sqrt{A^2 + 4d^2} - A\Big).} \tag{30}$$

For $d > 0$, any Einstein metric must therefore stay a positive absolute-energy distance above the Ricci-energy infimum. This is a consequence of the classical curvature inequalities, not a new nonexistence theorem.

More generally, if $0 \leqslant P/R \leqslant \alpha$, then

$$H \leqslant \alpha(C + H + J) + \sqrt{J(C + J)}. \tag{31}$$

For the negative Yau26 normalization $\delta_- = \|\mathrm{Ric}^\sharp + I\|_{\infty,\mathrm{op}} < 1$,

$$P \leqslant 4\delta_-^2 V, \quad R \geqslant 4(1 - \delta_-)^2 V, \quad \alpha = \frac{\delta_-^2}{(1 - \delta_-)^2} \tag{32}$$

is admissible.

Thus, at any fixed positive blow-up/handle ratio d/A, a pointwise near-Einstein sequence cannot simultaneously be a relative Ricci-energy- minimizing sequence. This is a usable restriction on a construction strategy. It does not exclude inputs whose energy stays above that minimum, and it does not decide whether the smooth manifold admits an Einstein metric elsewhere.

6. The resulting first-line picture

Three statements now have different, explicit roles.

1. The existing integral gap near the forbidden 2/3 topology uses the project's integral calibration proof. It remains a theorem about exclusion of near-Einstein metrics in its actual-monopole range.
2. The present theorem describes every sequence minimizing the classical *absolute* Ricci energy. It identifies the topological contribution to the relative defect and the independent scalar-coherence term.

3. The fixed-metric converse uses the earlier actual-form remainder and a scalar spectral gap. It starts from curvature-functional and period data, not from an assumed Ricci-small input. The scale-splitting example explains why the scalar-coherence hypothesis has content.

The prior actual-form supply uses $t = 1/2$, already an allowed negative Kähler–Einstein root of Round21. Substituting that value into $\operatorname{dist}(t, \{0, 1/2\})$ alone gives zero and proves no new obstruction. The additional conclusions here come from curvature-energy identities, actual topology, and control or removal of scalar variance.

The remaining Yau26 task is not concealed by these deductions: small integrated Ricci error must still be connected to the required pointwise input or to existence of an exact Einstein metric on the same smooth manifold. Neither follows merely from minimizing the classical energy. No new Yau26 branch is declared solved here.

Sources and proof dependence

- LeBrun, Ricci curvature and Seiberg–Witten theory, cited preprint version: Theorem 2.3, Proposition 3.2, Propositions 4.1–4.2, and Theorem 4.3. These supply the inequalities, class selection, and sharp infimum.
- Polarized application closure: project notation, metricwise class selection, and chamber details.
- Previous Kähler calculation: the Kähler-only minimizing-sequence profile, now extended above to all metrics with an explicit scalar-variance term.
- Actual-form remainder: needed for the complementary fixed-metric bridge, not for Theorem 1.
- The negative Yamabe representative uses the classical Yamabe theorem; all subsequent conformal monotonicity identities are proved in (26)–(29).